So far...

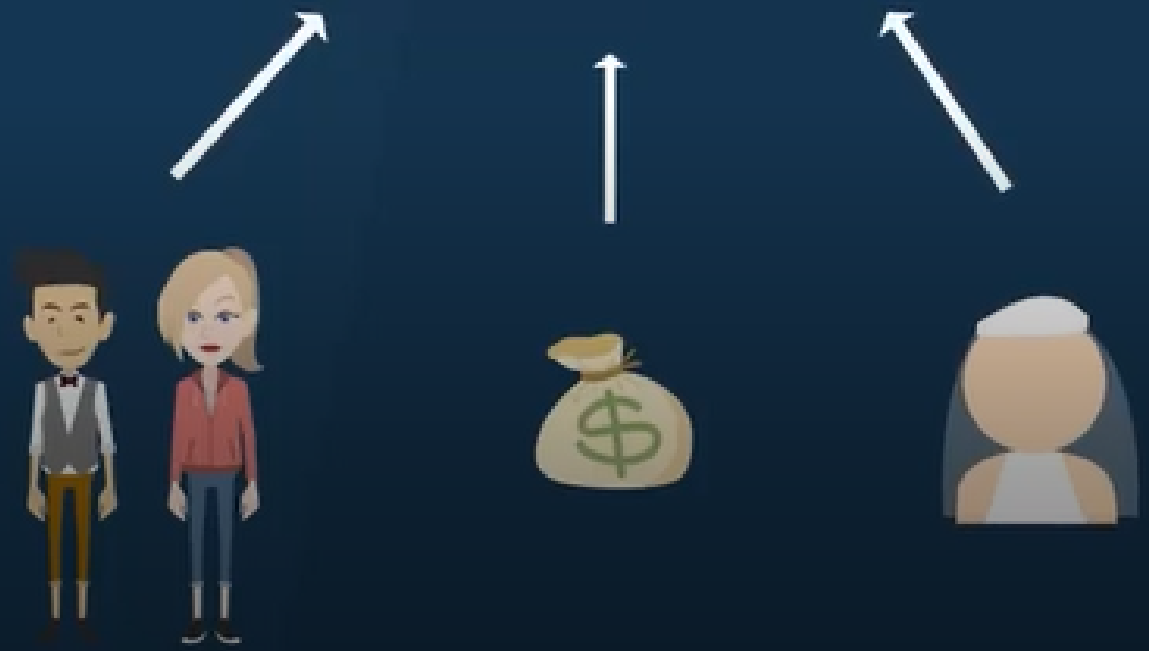
$$\hat{y}_i = b_0 + b_1 x_i$$

dependent variable

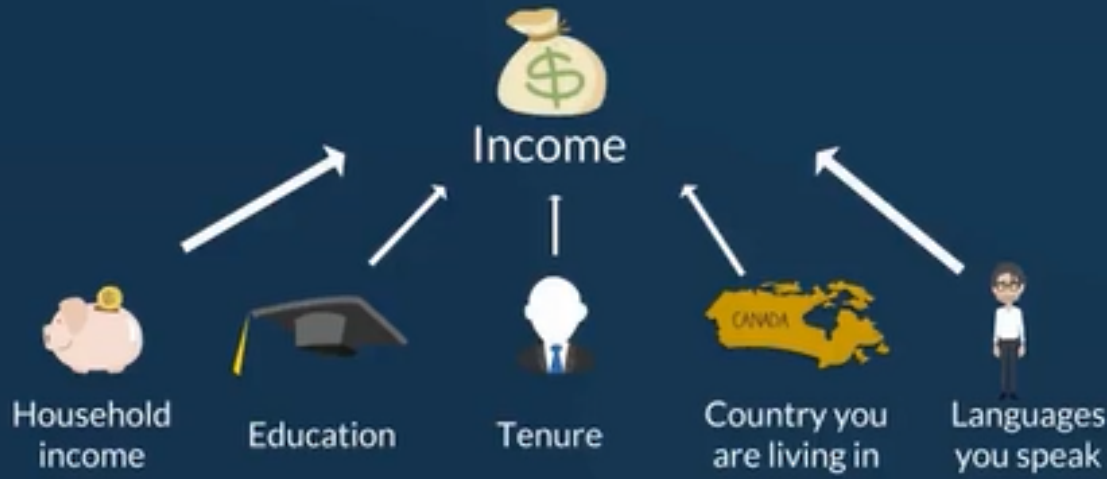
single regressor


House price
Location
Size
Year
calendar

College GPA = 0.48+0.0015*SAT

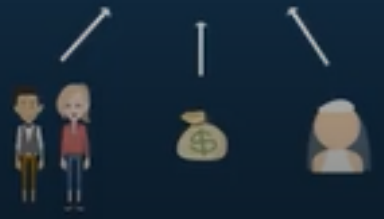


Good models require multivariate regressions, in order to address the higher complexity of problems





College GPA = 0.48+0.0015*SAT



Population model

$$\gamma = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

$$\gamma = \beta_0 + \beta_1 x_1 + \varepsilon$$

Multivariate regression equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

inferred value

intercept

Multivariate regression equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$



independent variable

independent variable

independent variable

Multivariate regression equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

coefficient coefficient coefficient

Multivariate regression equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

